3.4 A family $\{(M, g, \nabla^{-\alpha}, \nabla^{\alpha} = (\nabla^{-\alpha})^*)\}_{\alpha\in\mathbb{R}}$ of conjugate connection manifolds

For any pair (∇, ∇^*) of conjugate connections, we can define a 1-*parameter family of connections* $\{\nabla^{\alpha}\}_{\alpha\in\mathbb{R}}$, called the α-*connections* such that $(\nabla^{-\alpha}, \nabla^{\alpha})$ are dually coupled to the metric, with $\nabla^0 = \bar{\nabla} = {}^{\mathrm{LC}}\nabla$, $\nabla^1 = \nabla$ and $\nabla^{-1} = \nabla^*$. By observing that the scaled cubic tensor αC is also a totally symmetric cubic 3-covariant tensor, we can derive the α-connections from a statistical manifold (M, g, C) as:

$$\Gamma^{\alpha}_{ij,k} = \Gamma^0_{ij,k} - \frac{\alpha}{2} C_{ij,k}, \tag{45}$$

$$\Gamma^{-\alpha}_{ij,k} = \Gamma^0_{ij,k} + \frac{\alpha}{2} C_{ij,k}, \tag{46}$$

where $\Gamma^0_{ij,k}$ are the Levi-Civita Christoffel symbols, and $\Gamma_{ki,j} \overset{\Sigma}{=} \Gamma^l_{ij} g_{lk}$ (by index juggling).

The α-connection ∇^{α} can also be defined as follows:

$$g(\nabla^{\alpha}_X Y, Z) = g({}^{\mathrm{LC}}\nabla_X Y, Z) + \frac{\alpha}{2} C(X, Y, Z), \forall X, Y, Z \in \mathfrak{X}(M). \tag{47}$$

Theorem 2 (Family of information α-manifolds)**.** *For any $\alpha \in \mathbb{R}$, $(M, g, \nabla^{-\alpha}, \nabla^{\alpha} = (\nabla^{-\alpha})^*)$ is a conjugate connection manifold.*

The α-connections ∇^{α} can also be constructed directly from a pair (∇, ∇^*) of conjugate connections by taking the following weighted combination:

$$\Gamma^{\alpha}_{ij,k} = \frac{1+\alpha}{2} \Gamma_{ij,k} + \frac{1-\alpha}{2} \Gamma^*_{ij,k}. \tag{48}$$

3.5 The fundamental theorem of information geometry: ∇ κ-curved $\Leftrightarrow$ ∇^* κ-curved

We now state the fundamental theorem of information geometry and its corollaries:

Theorem 3 (Dually constant curvature manifolds)**.** *If a torsion-free affine connection ∇ has constant curvature κ then its conjugate torsion-free connection ∇^* has necessarily the* same *constant curvature κ.*

The proof is reported in [25] (Proposition 8.1.4, page 226).

A statistical manifold (M, g, C) is said α-*flat* if its induced α-connection is flat. It can be shown that $R^{\alpha} = -R^{-\alpha}$.

We get the following two corollaries:

Corollary 1 (Dually α-flat manifolds)**.** *A manifold $(M, g, \nabla^{-\alpha}, \nabla^{\alpha})$ is ∇^{α}-flat if and only if it is $\nabla^{-\alpha}$-flat.*

Corollary 2 (Dually flat manifolds ($\alpha = \pm 1$))**.** *A manifold (M, g, ∇, ∇^*) is ∇-flat if and only if it is ∇^*-flat.*

(See Theorem 3.3 of [9])

Let us now define the notion of constant curvature of a statistical structure [46]:

Definition 4 (Constant curvature κ)**.** *A statistical structure (M, g, ∇) is said of constant curvature κ when*

$$R^{\nabla}(X, Y)Z = \kappa\{g(Y, Z)X - g(X, Z)Y\}, \quad \forall X, Y, Z \in \Gamma(TM),$$

where $\Gamma(TM)$ denote the space of smooth vector fields.

It can be proved that the Riemann-Christoffel (RC) 4-tensors of conjugate α-connections [25] are related as follows:

$$g\left(R^{(\alpha)}(X, Y)Z, W\right) + g\left(Z, R^{(-\alpha)}(X, Y)W\right) = 0. \tag{49}$$

Thus we have $g\left(R^{\nabla^*}(X, Y)Z, W\right) = -g\left(Z, R^{\nabla}(X, Y)W\right)$.

Thus once we are given a pair of conjugate connections, we can always build a 1-parametric family of manifolds. Manifolds with constant curvature κ are interesting from the computational viewpoint as dual geodesics have simple closed-form expressions.

3.6 Conjugate connections from divergences: $(M, D) \equiv (M,\ {}^{D}g,\ {}^{D}\nabla,\ {}^{D}\nabla^* = {}^{D^*}\nabla)$

Loosely speaking, a divergence $D(\cdot : \cdot)$ is a smooth distance [138], potentially asymmetric. In order to define precisely a divergence, let us first introduce the following handy notations: $\partial_{i,\cdot} f(x,y) = \frac{\partial}{\partial x^i} f(x,y)$, $\partial_{\cdot,j} f(x,y) = \frac{\partial}{\partial y^j} f(x,y)$, $\partial_{ij,k} f(x,y) = \frac{\partial^2}{\partial x^i \partial x^j} \frac{\partial}{\partial y^k} f(x,y)$ and $\partial_{i,jk} f(x,y) = \frac{\partial}{\partial x^i} \frac{\partial^2}{\partial y^j \partial y^k} f(x,y)$, etc.

Definition 5 (Divergence)**.** *A divergence $D : M \times M \to [0, \infty)$ on a manifold M with respect to a local chart $\Theta \subset \mathbb{R}^D$ is a C^3-function satisfying the following properties:*

1. *$D(\theta : \theta') \geq 0$ for all $\theta, \theta' \in \Theta$ with equality holding iff $\theta = \theta'$ (law of the indiscernibles),*
2. *$\partial_{i,\cdot} D(\theta : \theta')|_{\theta=\theta'} = \partial_{\cdot,j} D(\theta : \theta')|_{\theta=\theta'} = 0$ for all $i, j \in [D]$,*
3. *$-\partial_{\cdot,i} \partial_{\cdot,j} D(\theta : \theta')|_{\theta=\theta'}$ is positive-definite.*

The *dual divergence* is defined by swapping the arguments:

$$D^*(\theta : \theta') := D(\theta' : \theta), \tag{50}$$

and is also called the *reverse divergence* (reference duality in information geometry). Reference duality of divergences is an involution: $(D^*)^* = D$.

The Euclidean distance is a metric distance but not a divergence. The squared Euclidean distance is a non-metric symmetric divergence. The metric tensor g yields Riemannian metric distance D_ρ but it is never a divergence.

From any given divergence D, we can define a conjugate connection manifold following the construction of Eguchi [42, 43] (1983):

Theorem 4 (Manifold from divergence)**.** *$(M, {}^{D}g, {}^{D}\nabla, {}^{D^*}\nabla)$ is an information manifold with:*

$$\begin{aligned} {}^{D}g &:= -\partial_{i,j} D(\theta : \theta')|_{\theta=\theta'} = {}^{D^*}g, && (51)\\ {}^{D}\Gamma_{ijk} &:= -\partial_{ij,k} D(\theta : \theta')|_{\theta=\theta'}, && (52)\\ {}^{D^*}\Gamma_{ijk} &:= -\partial_{k,ij} D(\theta : \theta')|_{\theta=\theta'}. && (53) \end{aligned}$$

The associated statistical manifold is $(M, {}^{D}g, {}^{D}C)$ with:

$${}^{D}C_{ijk} = {}^{D^*}\Gamma_{ijk} - {}^{D}\Gamma_{ijk}. \tag{54}$$

Since $\alpha {}^{D}C$ is a totally symmetric cubic tensor for any $\alpha \in \mathbb{R}$, we can derive a one-parameter family of conjugate connection manifolds:

$$\left\{ (M, {}^{D}g, {}^{D}C^{\alpha}) \equiv (M, {}^{D}g, {}^{D}\nabla^{-\alpha}, ({}^{D}\nabla^{-\alpha})^* = {}^{D}\nabla^{\alpha}) \right\}_{\alpha \in \mathbb{R}}. \tag{55}$$

In the remainder, we use the shortcut (M, D) to denote the divergence-induced information manifold $(M, {}^{D}g, {}^{D}\nabla, {}^{D}\nabla^*)$. Notice that it follows from construction that:

$${}^{D}\nabla^* = {}^{D^*}\nabla. \tag{56}$$

3.7 Dually flat manifolds (Bregman geometry): $(M, F) \equiv (M,\ {}^{B_F}g,\ {}^{B_F}\nabla,\ {}^{B_F}\nabla^* = {}^{B_{F^*}}\nabla)$

We consider dually flat manifolds that satisfy asymmetric Pythagorean theorems. These flat manifolds can be obtained from a canonical Bregman divergence.

Consider a *strictly convex smooth function* $F(\theta)$ called a *potential function*, with $\theta \in \Theta$ where Θ is an open convex domain. Notice that the function convexity does not change by an affine transformation. We associate to the potential function F a corresponding *Bregman divergence* (parameter divergence):

$$B_F(\theta : \theta') := F(\theta) - F(\theta') - (\theta - \theta')^\top \nabla F(\theta'). \tag{57}$$

We write also the Bregman divergence between point P and point Q as $D(P:Q) := B_F(\theta(P):\theta(Q))$, where $\theta(P)$ denotes the coordinates of a point P.

The information-geometric structure induced by a Bregman generator[9] is $(M, {}^F g, {}^F C) := (M, {}^{B_F} g, {}^{B_F} C)$ with:

$$
\begin{aligned}
{}^F g &:= {}^{B_F} g = -\left[\partial_i \partial_j B_F(\theta:\theta')|_{\theta'=\theta}\right] = \nabla^2 F(\theta), && (58)\\
{}^F \Gamma &:= {}^{B_F} \Gamma_{ij,k}(\theta) = 0, && (59)\\
{}^F C_{ijk} &:= {}^{B_F} C_{ijk} = \partial_i \partial_j \partial_k F(\theta). && (60)
\end{aligned}
$$

Since all coefficients of the Christoffel symbols vanish (Eq. 59), the information manifold is ${}^F\nabla$-*flat*. The Levi-Civita connection ${}^{\mathrm{LC}}\nabla$ is obtained from the metric tensor ${}^F g$ (usually not flat), and we get the conjugate connection $({}^F\nabla)^* = {}^F\nabla^1$ from $(M,\ {}^F g,\ {}^F C)$.

The Legendre-Fenchel transformation yields the *convex conjugate* F^* that is interpreted as the *dual potential function*:

$$F^*(\eta) := \sup_{\theta\in\Theta}\{\theta^\top \eta - F(\theta)\}. \tag{61}$$

Theorem 5 (Fenchel-Moreau biconjugation [52])**.** *If F is a lower semicontinuous*[10] *and convex function, then its Legendre-Fenchel transformation is involutive: $(F^*)^* = F$ (biconjugation).*

In a dually flat manifold, there exists two global dual affine coordinate systems $\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta)$, and therefore the manifold can be covered by a single chart. Thus if a probability family belongs to an exponential family then its natural parameters cannot belong to, say, a spherical space (that requires at least two charts).

We have the Crouzeix [32] identity relating the Hessians of the potential functions:

$$\nabla^2 F(\theta)\nabla^2 F^*(\eta) = I, \tag{62}$$

where I denote the $D\times D$ identity matrix. This Crouzeix identity reveals that $B = \{\partial_i\}_i$ and $B^* = \{\partial^j\}_j$ are the primal and reciprocal basis, respectively.

The Bregman divergence can be reinterpreted using Young-Fenchel (in)equality as the *canonical divergence* A_{F,F^*} [12]:

$$B_F(\theta:\theta') = A_{F,F^*}(\theta:\eta') = F(\theta) + F^*(\eta') - \theta^\top \eta' = A_{F^*,F}(\eta':\theta). \tag{63}$$

The *dual Bregman divergence* ${B_F}^*(\theta:\theta') := B_F(\theta':\theta) = B_{F^*}(\eta:\eta')$ yields

$$
\begin{aligned}
{}^F g^{ij}(\eta) &= \partial^i \partial^j F^*(\eta), \quad \partial^l :=: \frac{\partial}{\partial \eta^l} && (64)\\
{}^F {\Gamma^*}^{ijk}(\eta) &= 0, \quad {}^F C^{ijk} = \partial^i \partial^j \partial^k F^*(\eta) && (65)
\end{aligned}
$$

Thus the information manifold is both ${}^F\nabla$-flat and ${}^F\nabla^*$-flat: This structure is called a *dually flat manifold* (DFM). In a DFM, we have two global affine coordinate systems $\theta(\cdot)$ and $\eta(\cdot)$ related by the Legendre-Fenchel transformation of a pair of potential functions F and F^*. That is, $(M,F)\equiv(M,F^*)$, and the dual atlases are $\mathcal{A} = \{(M,\theta)\}$ and $\mathcal{A}^* = \{(M,\eta)\}$.

In a dually flat manifold, any pair of points P and Q can either be linked using the ∇-geodesic (that is θ-straight) or the ∇^*-geodesic (that is η-straight). In general, there are $2^3 = 8$ types of *geodesic triangles* in a dually flat manifold.

[9] Here, we define a Bregman generator as a proper, lower semi-continuous, and strictly convex and C^3 differentiable real-valued function.

[10] A function f is lower semicontinous (lsc) at x_0 iff $f(x_0) \le \lim_{x\to x_0}\inf f(x)$. A function f is lsc if it is lsc at x for all x in the function domain.